Shourya Bhardwaj

AI systems, automation & experimental engineering

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Builder focused on local-first AI agents, automation, and biologically inspired simulation. I work from architecture through implementation, debugging, validation, and iteration to turn ambitious ideas into working systems.

Selected Engineering Projects

Nova

In development

Primary builder · AI Agents / Systems Engineering

- Architected a C++20 runtime and Python voice pipeline, integrating OpenWakeWord and faster-whisper with request normalization, routing, and multi-step planning.
- Implemented plan validation and filesystem authorization to reject unresolved operations or context references before execution and return structured failures.
- Integrated Playwright browser operations and observation-based execution guards that stop repeated failures and browser actions producing no observable state change.
- Defined CMake test targets for decomposition, primitive mapping, filesystem policies, and runtime integration; kept execution results and context available for debugging.

github.com/shoyo-nova92/Nova-AI

Neurofly

Experimental exploration

Project builder · Computational Neuroscience / Simulation

- Designing an experimental Drosophila-inspired olfactory simulation, mapping odor concentration fields through receptor activation and neural response toward autonomous movement.
- Exploring bilateral sensing and competing odor sources in a virtual arena; treating an ESP32 robot as a future embodiment direction, not completed integration.

Freelance & Creative Experience

Independent design & editing

Independent visual design and video-editing work across promotional and product communication. Selected compositions include LinkScope, Identix, Clarion AI, and Spectra AI.

View selected creative work ↗

Technical Activity

Hands-on Codex workshop

Technical workshop · Oct 2026

- Attended a hands-on workshop on structured prompting, agent workflows, project guidance, web development, and testing.

Technical Capabilities

AI & agent systems

Planning and tool use · Validation and execution workflows · Voice-input pipelines · Browser and desktop automation

Applied interaction

faster-whisper · OpenWakeWord · Playwright · Input normalization and routing

Creative

Visual design · Video editing · Visual communication

Systems & software

C++ · Python · CMake · Runtime architecture · Filesystem operations · Debugging and testing

Research & simulation

Biologically inspired control · Olfactory simulation · Neural navigation exploration